Table 10-1. Typical Failure Modes of Centrifugal Pump Assemblies

FAILURE MODE	FAILURE CAUSE	FAILURE EFFECT
Reduction in suction head	- Pump cavitation	- Loss of pump efficiency
Reduction in pump pressure	- Pump cavitation	- Eventual erosion of impeller, casing - Pump noise and vibration
Component corrosion	- Incorrect fluid - Excessive flow rate for fluid	- Eventual catastrophic pump failure
Shaft deflection	- High radial thrust on pump Rotor	- Eventual shaft and pump failure
Shaft unbalance	- Impeller wear	- Shaft deflection and Misalignment - Stuffing box leakage - Seal leakage - Bearing wear
Air leak thru gasket / stuffing box	- Damaged gasket	- Loss of pump head
External Leakage	- Seal failure - Worn mechanical seal - Scored shaft sleeve - Stuffing box improperly packed	- Depends on type of fluid and criticality as to time of failure
Mechanical noise	- Debris in the impeller - Impeller out of balance - Bent shaft - Worn/damaged bearing - Foundation not rigid - Cavitation	- Eventual wear of impeller and other components
Positive suction head too low	- Clogged suction pipe - Valve on suction line only partially open	- Suction cavitation - Noisy operation - Low discharge pressure - High output flow rate
Pump discharge head too high	- Clogged discharge pipe - Discharge line valve only partially open	- Discharge cavitation - Noisy operation - Low output flow rate
Suction line / impeller clogged	- Contaminants	- Loss of pump output / reduced flow
Worn / broken impeller	- Wrong flow rate, contaminants	- Loss of pump output / reduced flow
Thrust bearing failure	- Excessive axial load	- Pump failure